

9/22 ch.1.6

Cauchy Schwartz Inequality: $|\vec{u}| |\vec{v}| \geq |\vec{u} \cdot \vec{v}|$

$$\frac{|\vec{u} \cdot \vec{v}|}{|\vec{u}| |\vec{v}|} \leq 1$$

Orthogonal: 互相垂直 · 正交

Orthonormal: two vectors 互相垂直且長度為1

Distance between points

DEFINITION

Let $\mathbf{x} = (x_1, \dots, x_n)$ and $\mathbf{y} = (y_1, \dots, y_n)$ be two points in $\mathbb{R}^n$. The distance between $\mathbf{x}$ and $\mathbf{y}$ is denoted $d(\mathbf{x}, \mathbf{y})$ and is defined by

$$d(\mathbf{x}, \mathbf{y}) = \sqrt{(x_1 - y_1)^2 + \dots + (x_n - y_n)^2}$$

Note: We can also write this distance formula as

$$d(\mathbf{x}, \mathbf{y}) = \|\mathbf{x} - \mathbf{y}\|$$

Let $\mathbf{u}$ and $\mathbf{v}$ be vectors in $\mathbb{R}^n$.

(a) **Triangle inequality:** $\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|$

This inequality tells us that the length of one side of a triangle cannot exceed the sum of the lengths of the other two sides. See Figure 1.24(a).

(b) **Pythagorean Theorem:** If $\mathbf{u} \cdot \mathbf{v} = 0$, then $\|\mathbf{u} + \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2$.